

Weighted tensor products of Joyal species, graphs, and charades

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Abstract

Motivated by the weighted Hurwitz product on sequences in an algebra, we produce a family of monoidal structures on the category of Joyal species. We suggest a family of tensor products for charades. We begin by seeing weighted differential algebras and weighted Rota-Baxter algebras as monoids for a monoidal structure on the category of graphs in a monoidal additive category.

Contents

1	Introduction	2
2	Review of λ-weighted derivations and Rota-Baxter operators	2
3	Graphs in monoidal additive categories	6
4	The L-tensor product of species	10
5	A combinatorial interpretation	12

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6	The iterated tensor and coherence	12
7	Promonoidal structures on $\mathfrak{S}$	14
8	The weighted bimonoidale structure on $\text{fam}\mathfrak{S}$	15
9	Weighted categorical derivations	16
10	The iterated tensor product again	18
11	Tensor products for charades	19
12	The dimension sequence	22
13	Weighted categorical Rota-Baxter operators	23

1 Introduction

There are monoidal structures on the category $\text{Gph}\mathcal{V}$ graphs in a monoidal additive category $\mathcal{V}$ for which weighted differential monoids and weighted Rota-Baxter monoids (see [10], for example) can both be seen as monoids. We also produce a family of monoidal structures on the category of Joyal species. In particular, this defines an interesting family of tensor products for linear representations of the symmetric groups. We also suggest a family of tensor products for charades (see [8, 21]) which generalizes, in particular, the essentially classical tensor product of representations of the general linear groups over a finite field, proved braided in [16].

2 Review of λ -weighted derivations and Rota-Baxter operators

The inspiration for the family of tensor products on species came from the λ -weighted product of Hurwitz series as discussed in [10, 11] and their references. They begin by defining a *derivation of weight* λ on an algebra A over a commutative ring k , with given $\lambda \in k$, to be a k -module morphism $d: A \rightarrow A$ satisfying

$$d(ab) = d(a)b + ad(b) + \lambda d(a)d(b) .$$

They note the generalized Leibnitz rule

$$d^n(ab) = \sum_{k=0}^n \binom{n}{k} \binom{n-k}{j} \lambda^k d^{n-j}(a) d^{k+j}(b) .$$

However, I prefer to write this in the form

$$d^n(ab) = \sum_{n=r+s+t} \binom{n}{r, s, t} \lambda^t d^{r+t}(a) d^{s+t}(b) \quad (2.1)$$

to emphasise the relationship to the trinomial expansion rule for $(x + y + \lambda xy)^n$. Here

$$\binom{n}{r, s, t} = \frac{n!}{r!s!t!}.$$

The λ -Hurwitz product on $A^{\mathbb{N}}$ can similarly be defined by

$$(f \cdot^\lambda g)(n) = \sum_{n=r+s+t} \binom{n}{r, s, t} \lambda^t f(r+t) g(s+t). \quad (2.2)$$

Example 1. For $\lambda = 0$, $k = \mathbb{R}$ and A the algebra of smooth functions $f: \mathbb{R} \rightarrow \mathbb{R}$ under pointwise addition and multiplication, the differentiation function $d: A \rightarrow A$ is a 0-weighted derivation by the classical Leibnitz rule.

Example 2. For λ invertible, $k = \mathbb{R}$ and A the algebra of functions $f: \mathbb{R} \rightarrow \mathbb{R}$ under pointwise addition and multiplication, the function $d: A \rightarrow A$ defined by

$$d(f)(x) = \frac{f(x + \lambda) - f(x)}{\lambda}$$

is a λ -weighted derivation.

Example 3. Define $d: A^{\mathbb{N}} \rightarrow A^{\mathbb{N}}$ by $d(s)(n) = s(n+1) - s(n)$. This d is a 1-weighted derivation when $A^{\mathbb{N}}$ is equipped with the pointwise addition and multiplication.

Example 4. Define $d: A^{\mathbb{N}} \rightarrow A^{\mathbb{N}}$ by $d(f)(n) = f(n+1)$. This d is a λ -weighted derivation when $A^{\mathbb{N}}$ is equipped with the λ -Hurwitz product for any λ . Notice that we have an algebra morphism $d^*: A^{\mathbb{N}} \rightarrow (A^{\mathbb{N}})^{\mathbb{N}}$ defined by $d^*(f)(m)(n) = f(m+n)$. This may motivate the next definition.

Define $d^*: A \rightarrow A^{\mathbb{N}}$ by $d^*(a)(n) = d^n(a)$. We see that the Leibnitz rule (2.1) amounts to:

Proposition 5. $d^*: A \rightarrow A^{\mathbb{N}}$ is an algebra morphism for all λ -weighted derivations d on A , where $A^{\mathbb{N}}$ has the λ -Hurwitz product.

In fact, $A \mapsto A^{\mathbb{N}}$ is a comonad

$$G = \left((-)^{\mathbb{N}}, \varepsilon, \delta \right) \quad (2.3)$$

on the category Alg_k of k -algebras whose Eilenberg-Moore-coalgebras are k -algebras A equipped with a λ -derivation, so-called λ -derivation algebras; write DA_λ for the category of these. The morphism $d^*: A \rightarrow A^{\mathbb{N}}$ is the coaction of the comonad.

Where there is differentiation, there should also be integration. A *Rota-Baxter operator of weight λ* on a k -algebra A is a k -linear morphism $P: A \rightarrow A$ satisfying

$$P(a)P(b) = P(P(a)b + P(a)b + \lambda ab) . \quad (2.4)$$

The pair (A, P) is called a λ -weighted Rota-Baxter algebra. Write RBA_λ for the category of these.

Example 6. For $\lambda = 0$, $k = \mathbb{R}$ and A the algebra of continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ under pointwise addition and multiplication, the integration function $P: A \rightarrow A$, defined by $P(f)(x) = \int_0^x f(t)dt$, is a 0-weighted Rota-Baxter operator by the classical integration-by-parts rule.

Example 7. For $\lambda = 1$ and any k -algebra A , define $P: A^\mathbb{N} \rightarrow A^\mathbb{N}$ to take a sequence u in A to its sequence $P(u)$ of partial sums:

$$P(u)(n) = \sum_{i=0}^{n-1} u(i) .$$

Then P is a 1-weighted Rota-Baxter operator on $A^\mathbb{N}$ with pointwise addition and multiplication. See [1, 20]. For d the consecutive difference operator as defined in Example 3, notice that $d \circ P = 1_{A^\mathbb{N}}$.

Example 8. If Q is a 1-weighted Rota-Baxter operator on A then $P(a) = \lambda Q(a)$ defines a λ -weighted Rota-Baxter operator P on A .

A λ -weighted derivation RB-algebra is a k -algebra A equipped with a λ -weighted derivation d and a λ -weighted Rota-Baxter operator P such that $d \circ P = 1_A$. Write DRB_λ for the category of these.

Proposition 9. [See [10]] Let P be a RB-operator of weight λ on A . Then $A^\mathbb{N}$ equipped with the λ -Hurwitz product, the derivation d of Example 4, and P defined by

$$P(f)(n) = \begin{cases} P(f(0)) & \text{for } n = 0 \\ f(n-1) & \text{for } n > 0 \end{cases}$$

is a λ -weighted derivation RB-algebra. Moreover, the following square commutes.

$$\begin{array}{ccc} A^\mathbb{N} & \xrightarrow{P} & A^\mathbb{N} \\ \text{ev}_0 \downarrow & & \downarrow \text{ev}_0 \\ A & \xrightarrow{P} & A \end{array}$$

With a little more work following Proposition 9, we see that the comonad G (2.3) lifts to $\mathbf{RBA}_\lambda$. In particular, with V denoting the forgetful functor, we have a comonad $\bar{G}$ and a commutative square

$$\begin{array}{ccc} \mathbf{RBA}_\lambda & \xrightarrow{\bar{G}} & \mathbf{RBA}_\lambda \\ V \downarrow & & \downarrow V \\ \mathbf{Alg}_k & \xrightarrow{G} & \mathbf{Alg}_k \end{array} .$$

Write $\mathbf{RBA}_{\lambda-}$ for the category of λ -weighted Rota-Baxter algebras where we do not insist on the algebras having a unit.

Proposition 10. *For each $(A, \cdot, P) \in \mathbf{RBA}_{\lambda-}$, there is an associative binary operation $a \diamond b$ defined on A by*

$$a \diamond b = P(a) \cdot b + a \cdot P(b) + \lambda a \cdot b .$$

Moreover, $T(A, \cdot, P) = (A, \diamond, P)$ defines an endofunctor

$$T: \mathbf{RBA}_{\lambda-} \longrightarrow \mathbf{RBA}_{\lambda-} ,$$

which is (well-)copointed by a natural transformation $\gamma: T \Rightarrow 1_{\mathbf{RBA}_{\lambda-}}$ whose component at $(A, \cdot, P)$ is $P: (A, \diamond, P) \rightarrow (A, \cdot, P)$.

Remark 11. The day after my seminar talk of 4 February 2015, on the material of this section and Section 4, Stephen Lack made the following comments.

1. Consider the category $[\Sigma\mathbb{N}, \mathcal{V}]$ whose objects are pairs (M, d) consisting of an object M of a nice monoidal additive category $\mathcal{V}$ and an endomorphism $d: M \rightarrow M$. The forgetful functor

$$U: [\Sigma\mathbb{N}, \mathcal{V}] \longrightarrow \mathcal{V} , \tag{2.5}$$

taking (M, d) to M , has a right adjoint taking M to $(M^\mathbb{N}, d)$ where $d(f)(n) = f(n+1)$. There is a monoidal structure on $[\Sigma\mathbb{N}, \mathcal{V}]$ defined by

$$(M, d) \otimes^\lambda (N, d) = (M \otimes N, d \otimes 1 + 1 \otimes d + \lambda d \otimes d) .$$

The monoids in this monoidal category are precisely λ -derivation algebras. Moreover, U (2.5) and its right adjoint form a monoidal adjunction which therefore passes to the categories of monoids. This adjunction generates the comonad G (2.3) on the category $\mathbf{Mon}\mathcal{V}$ of monoids in $\mathcal{V}$.

2. There is a bialgebra structure on the polynomial algebra $k[x]$ with comultiplication the algebra morphism $\delta: k[x] \rightarrow k[x, y] \cong k[x] \otimes k[x]$ defined by

$$\delta(x) = x + y + \lambda xy .$$

Then the convolution product on the left-hand side of the canonical isomorphism

$$\mathrm{Mod}_k(k[x], A) \cong A^{\mathbb{N}}$$

transports to the λ -Hurwitz product on $A^{\mathbb{N}}$.

3. It feels like there should be a multicategory/promonoidal/substitute structure on $[\Sigma\mathbb{N}, \mathcal{V}]$ for dealing with RB-algebras.

3 Graphs in monoidal additive categories

Let $\mathcal{V}$ be a monoidal additive category. We act as if the monoidal structure were strict.

Let $\mathrm{Gph}\mathcal{V}$ be the category of directed graphs in $\mathcal{V}$. So an object has the form of a pair of parallel morphisms $s, t: E \rightarrow A$ in $\mathcal{V}$; we use s and t for source and target morphisms in all graphs. A morphism $(f, \phi): (A, E) \rightarrow (B, F)$ in $\mathrm{Gph}\mathcal{V}$ consists of morphisms f and ϕ making the following diagram commute.

$$\begin{array}{ccccc} A & \xleftarrow{s} & E & \xrightarrow{t} & A \\ f \downarrow & & \downarrow \phi & & \downarrow f \\ B & \xleftarrow{s} & F & \xrightarrow{t} & B \end{array}$$

Write $\mathrm{ver}: \mathrm{Gph}\mathcal{V} \rightarrow \mathcal{V}$ for the forgetful functor taking (A, E) to A and write $\mathrm{edg}: \mathrm{Gph}\mathcal{V} \rightarrow \mathcal{V}$ for the forgetful functor taking (A, E) to E .

We will use the notation $\langle n \rangle = \{1, 2, \dots, n\}$. For $R \subseteq \langle n \rangle$, write

$$\chi_R: \langle n \rangle \rightarrow \{s, t\}$$

for the characteristic function of R defined by

$$\chi_R(i) = \begin{cases} s & \text{for } i \in R \\ t & \text{for } i \notin R. \end{cases}$$

Choose an endomorphism $\lambda: I \rightarrow I$ of the tensor unit I in $\mathcal{V}$. For any $f: A \rightarrow B$ in $\mathcal{V}$, we define $(\lambda f: A \rightarrow B) = (\lambda \otimes f: I \otimes A \rightarrow I \otimes B)$.

Given a list $(A_1, E_1), \dots, (A_n, E_n)$ of objects of $\mathrm{Gph}\mathcal{V}$, we define an n -fold tensor product

$$\otimes_{1 \leq i \leq n}^{\lambda} (A_i, E_i) = (\otimes_{1 \leq i \leq n} A_i, \otimes_{1 \leq i \leq n} E_i), \quad (3.6)$$

where

$$s = \sum_{\emptyset \neq R \subseteq \langle n \rangle} \lambda^{(\#R-1)} \chi_R(1) \otimes \dots \otimes \chi_R(n) \quad \text{and} \quad t = t \otimes \dots \otimes t. \quad (3.7)$$

For $n = 2$ this gives a binary tensor product

$$(A, E) \otimes^\lambda (B, F) = (A \otimes B, E \otimes F)$$

with

$$s = \lambda s \otimes s + s \otimes t + t \otimes s \quad \text{and} \quad t = t \otimes t .$$

The unit for this tensor is the graph (I, I) with $s = 0: I \rightarrow I$ and $t = 1_I: I \rightarrow I$.

Proposition 12. *A monoidal structure on $\text{Gph}\mathcal{V}$ is defined by (3.6) for any given $\lambda \in \mathcal{V}(I, I)$. Both ver and $\text{edg}: \text{Gph}\mathcal{V} \rightarrow \mathcal{V}$ are strict monoidal.*

Proof. Easy calculations of the source morphisms for

$$\left((A, E) \otimes^\lambda (B, F) \right) \otimes^\lambda (C, G) \quad \text{and} \quad (A, E) \otimes^\lambda \left((B, F) \otimes^\lambda (C, G) \right)$$

show they agree with that of the triple tensor product. What this really means is that the associativity constraints for $\mathcal{V}$ lift through ver and edg to $\text{Gph}\mathcal{V}$ and are therefore coherent. $\square$

Let $[\Sigma\mathbb{N}, \mathcal{V}]$ denote the category whose objects $(A, e: A \rightarrow A)$ consist of an object A of $\mathcal{V}$ equipped with an endomorphism e . Let

$$J: [\Sigma\mathbb{N}, \mathcal{V}] \rightarrow \text{Gph}\mathcal{V} \tag{3.8}$$

be the functor defined by $J(A, e) = (A, A)$ with $s = e$ and $t = 1_A$; and $Jf = (f, f)$. Notice also that a morphism $(f, \phi): (B, F) \rightarrow J(A, e)$ in $\text{Gph}\mathcal{V}$ with codomain in the subcategory amounts to a commutative diagram

$$\begin{array}{ccccc} F & \xrightarrow{t} & B & \xrightarrow{f} & A \\ s \downarrow & & & & \downarrow e \\ B & \xrightarrow{f} & & & A \end{array} \tag{3.9}$$

where ϕ is forced to be $f \circ t: F \rightarrow A$. Clearly J is fully faithful and the monoidal structure of Proposition 12 restricts to a monoidal structure on $[\Sigma\mathbb{N}, \mathcal{V}]$ yielding (3.8) as a strict monoidal functor.

Definition 13. A λ -weighted derivation monoid in $\mathcal{V}$ is a monoid (A, d) in $[\Sigma\mathbb{N}, \mathcal{V}]$ equipped with the monoidal structure obtained as the restriction through (3.8) of that of Proposition 12 on $\text{Gph}\mathcal{V}$.

More explicitly, a λ -weighted differential monoid is a monoid A in $\mathcal{V}$ equipped with an endomorphism $d: A \rightarrow A$ satisfying the λ -weighted equation:

$$d \circ \mu = \mu \circ (\lambda d \otimes d + d \otimes 1 + 1 \otimes d) . \tag{3.10}$$

There is an isomorphism of categories

$$\text{op}: \text{Gph}\mathcal{V} \longrightarrow \text{Gph}\mathcal{V} \quad (3.11)$$

taking (A, E) to $(A, E)^{\text{op}}$ for which A and E are unchanged but s and t have been interchanged.

Put

$$J^{\text{op}} = \left([\Sigma\mathbb{N}, \mathcal{V}] \xrightarrow{J} \text{Gph}\mathcal{V} \xrightarrow{\text{op}} \text{Gph}\mathcal{V} \right) . \quad (3.12)$$

Like J , this composite J^{op} is fully faithful.

However, the image of J^{op} is **not** closed under the monoidal structure of Proposition 12. All we obtain on $[\Sigma\mathbb{N}, \mathcal{V}]$ is a multicategory structure. The sets of multimorphisms are defined by

$$\text{P}_\lambda((A_1, p_1), \dots, (A_n, p_n); (B, p)) = \text{Gph}\mathcal{V} \left(\otimes_{1 \leq i \leq n}^{\lambda} J^{\text{op}}(A_i, p_i), J^{\text{op}}(B, p) \right) . \quad (3.13)$$

To be more explicitly, for $R \subseteq \langle n \rangle$, write

$$\chi'_R: \langle n \rangle \longrightarrow \{1, p_1, \dots, p_n\}$$

for the function defined by

$$\chi'_R(i) = \begin{cases} 1 & \text{for } i \in R \\ p_i & \text{for } i \notin R . \end{cases}$$

Then, an element of the set (3.13), a *multimorphism*, is a morphism

$$f: A_1 \otimes \dots \otimes A_n \longrightarrow B$$

satisfying the equation

$$f \circ (p_1 \otimes \dots \otimes p_n) = p \circ f \circ \sum_{\emptyset \neq R \subseteq \langle n \rangle} \lambda^{(\#R-1)} \chi'_R(1) \otimes \dots \otimes \chi'_R(n) .$$

Definition 14. A λ -weighted Rota-Baxter monoid in $\mathcal{V}$ is an object (A, p) of $[\Sigma\mathbb{N}, \mathcal{V}]$ (that is, $p: A \rightarrow A$ in $\mathcal{V}$) equipped with the structure of monoid on $J^{\text{op}}(A, p)$ in the monoidal category $\text{Gph}\mathcal{V}$ of Proposition 12.

This definition should make the calculation of free weighted Rota-Baxter monoids possible; compare [20, 2] for $\lambda = 1, -1$.

To make Definition 14 a little more explicit, as expected, a λ -weighted Rota-Baxter monoid (A, p) is a monoid A in $\mathcal{V}$ equipped with an endomorphism $p: A \rightarrow A$ satisfying (3.14):

$$\mu \circ (p \otimes p) = p \circ \mu \circ (\lambda 1 \otimes 1 + 1 \otimes p + p \otimes 1) . \quad (3.14)$$

Derivations and Rota-Baxter operators are not the only sources of monoids for the monoidal structure of Proposition 12. The forgetful functor

$$\text{ve}: \text{Gph}\mathcal{V} \longrightarrow \mathcal{V} \otimes \mathcal{V}, \quad (3.15)$$

taking the graph (A, E) to the pair (A, E) , is strict monoidal and has a right adjoint R defined by

$$R(X, Y) = (X, X \oplus X \oplus Y) \quad (3.16)$$

with $s = \text{pr}_1$ (the first projection) and $t = \text{pr}_2$ (the second projection). It follows that R is monoidal and hence takes monoids to monoids.

Example 15. Take $\mathcal{V} = \text{Mod}_k$, the category of modules over a commutative ring k . For a graph (A, E) in this $\mathcal{V}$, we can write $e: a \rightarrow b$ to mean $a, b \in A$, $e \in E$ with $s(e) = a, t(e) = b$. For k -algebras A and B , we obtain a monoid $R(A, B)$ in $\text{Gph}\mathcal{V}$: the graph is $\text{pr}_1, \text{pr}_2: A \oplus A \oplus B \rightarrow A$ and the multiplication is defined by:

$$\begin{aligned} & ((a_1, a_2, b): a_1 \rightarrow a_2) \cdot ((c_1, c_2, d): c_1 \rightarrow c_2) \\ &= (\lambda a_1 c_1 + a_1 c_2 + a_2 c_1, a_2 c_2, bd): a_1 c_1 \rightarrow a_2 c_2 \end{aligned}$$

Of course the $\mathcal{V}$ -functor J (3.8) has both adjoints if $\mathcal{V}$ is complete and cocomplete enough. In particular, the right adjoint

$$K: \text{Gph}\mathcal{V} \longrightarrow [\Sigma\mathbb{N}, \mathcal{V}] \quad (3.17)$$

is defined by taking $K(A, E)$ to be the equalizer of the two morphisms

$$s^{\mathbb{N}}, t^{\text{succ}}: E^{\mathbb{N}} \rightarrow A^{\mathbb{N}}$$

equipped with the endomorphism $e: K(A, E) \rightarrow K(A, E)$ induced by E^{succ} . Here $\text{succ}: \mathbb{N} \rightarrow \mathbb{N}$ is the successor function $n \mapsto n + 1$. Since J (3.8) is strong monoidal for the monoidal structures under discussion, the adjunction $J \dashv K$ is monoidal. So K takes monoids to monoids.

In particular, if (A, p) is a λ -weighted Rota-Baxter monoid in $\mathcal{V}$, then K takes the graph (A, A) with $s = 1_A$ and $t = p$ to a λ -weighted derivational monoid in $\mathcal{V}$. The underlying monoid is the limit of the diagram

$$A \xleftarrow{p} A \xleftarrow{p} A \xleftarrow{p} \dots$$

of monoids in $\mathcal{V}$.

Example 16. Taking $\mathcal{V} = \text{Vect}_k$ and a λ -weighted Rota-Baxter k -algebra $p: A \rightarrow A$, we have the λ -weighted derivational k -algebra

$$K(J(A, p)^{\text{op}}) = \{a \in A^{\mathbb{N}} \mid p(a_{n+1}) = a_n\}$$

with $d(a)_n = a_{n+1}$. Moreover, $K(J(A, p)^{\text{op}})$ supports a λ -weighted Rota-Baxter operator p defined by $p(a)_n = p(a_n)$. Notice too that $d \circ p = 1$.

We conclude this section by describing the promonoidal structure in the sense of Day [4] with respect to which the monoidal structure of Proposition 12 is convolution.

Let $\mathbb{G}$ denote the category whose only objects are 0 and 1, with the only non-identity morphisms $\sigma, \tau: 1 \rightarrow 0$. Write $I_*\mathbb{G}$ for the free $\mathcal{V}$ -category on $\mathbb{G}$. Then $\text{Gph}\mathcal{V} = [\mathbb{G}, \mathcal{V}] = [I_*\mathbb{G}, \mathcal{V}]$ where the first set of square brackets means the ordinary functor category while the second means the $\mathcal{V}$ -enriched functor category. The promonoidal structure in question is technically on $I_*\mathbb{G}$ in the $\mathcal{V}$ -enriched sense. However, we can look at it as consisting of an ordinary a functor

$$P: \mathbb{G}^{\text{op}} \times \mathbb{G}^{\text{op}} \longrightarrow \text{Gph}\mathcal{V} \quad (3.18)$$

and an object $J \in \text{Gph}\mathcal{V}$. Of course J is just the graph $0, 1: I \rightarrow I$ which is the tensor unit. We can regard P as a “cograph of cographs of graphs” (although a cograph looks just like a graph):

$$\begin{array}{ccc} I & \xrightarrow{(1,0)} & I \oplus I = 2 \cdot I \\ \downarrow (1,0) & \searrow (0,1) & \downarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\ 2 \cdot I = I \oplus I & \xrightarrow{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}} & ((\lambda, 1, 1, 0), (0, 0, 0, 1)): I \rightarrow 4 \cdot I \\ & \searrow \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \end{array}$$

4 The L -tensor product of species

Let $\mathfrak{S}$ denote the groupoid whose objects are finite sets and whose morphisms are bijective functions. We write $U + V$ for the disjoint union of sets U and V ; this is the binary coproduct as objects of the category Set of sets and all functions. It is not the coproduct in $\mathfrak{S}$; yet it does provide the symmetric monoidal structure on $\mathfrak{S}$ of interest here. When we write $X = A + B$ for A and B subsets of a set X , we mean $X = A \cup B$ and $\emptyset = A \cap B$.

We have the particular finite sets $\langle n \rangle = \{1, 2, \dots, n\}$.

Let $\mathcal{V}$ denote a monoidal category with finite coproducts which are preserved by tensoring on either side by an object. The tensor product of $V, W \in \mathcal{V}$ is denoted by $V \otimes W$ and the unit object by I . Justified by coherence theorems (see [15] for example), we write as if the monoidal structure on $\mathcal{V}$ were strictly associative and strictly unital. For any set S , write $S \cdot V$ for the coproduct of S copies of $V \in \mathcal{V}$, when it exists (as it does for S finite).

The *category of $\mathcal{V}$ -valued Joyal species*, after [12, 13], is the functor category $[\mathfrak{S}, \mathcal{V}]$. The objects will simply be called *species* when $\mathcal{V}$ is understood.

Suppose $L: \mathfrak{S} \rightarrow \mathcal{Z}\mathcal{V}$ is a braided strong monoidal functor into the monoidal centre (in the sense of [14]) of $\mathcal{V}$. We have natural isomorphisms

$$u_{X,V}: LX \otimes V \cong V \otimes LX$$

such that

$$\begin{array}{ccc} LX \otimes V \otimes W & \xrightarrow{u_{X,V} \otimes 1_W} & V \otimes LX \otimes W \\ & \searrow u_{X,V \otimes W} \quad \swarrow 1_V \otimes u_{X,W} & \\ & V \otimes W \otimes LX & \end{array} .$$

If $\mathcal{V}$ itself is braided (*a fortiori* symmetric), we can take a braided strong monoidal functor $L: \mathfrak{S} \rightarrow \mathcal{V}$ since then there is a canonical braided strong monoidal functor $\mathcal{V} \rightarrow \mathcal{Z}\mathcal{V}$.

By way of example, we could have any finite set Λ and $LX = \Lambda^X \cdot I$ with $L\sigma = \Lambda^{\sigma^{-1}} \cdot I$ for any bijective function σ .

Define the *L-tensor product* $F \otimes^L G$ of species F and G on objects $X \in \mathfrak{S}$ by

$$(F \otimes^L G)X = \sum_{X=U \cup V} L(U \cap V) \otimes FU \otimes GV . \quad (4.19)$$

The definition of $F \otimes^L G$ on morphisms is clear since any bijective function $\sigma: X \rightarrow Y$ restricts to bijections

$$U \rightarrow \sigma U, \quad V \rightarrow \sigma V, \quad U \cup V \rightarrow \sigma U \cup \sigma V, \quad U \cap V \rightarrow \sigma U \cap \sigma V .$$

Let $J: \mathfrak{S} \rightarrow \mathcal{V}$ be the species whose value at X is the unit I for tensor in $\mathcal{V}$ when X is empty and is initial in $\mathcal{V}$ otherwise. Clearly J is a unit for the L -tensor product in the sense that we have canonical isomorphisms

$$\lambda_G: J \otimes^L G \rightarrow G \quad \text{and} \quad \rho_F: F \rightarrow F \otimes^L J .$$

Associativity isomorphisms

$$\alpha_{F,G,H}: (F \otimes^L G) \otimes^L H \cong F \otimes^L (G \otimes^L H) \quad (4.20)$$

are obtained using the following fact easily proved by Venn diagrams.

Lemma 17. $(U \cup V) \cap W + U \cap V \cong U \cap (V \cup W) + V \cap W$

In the case $L = J$, we recover from (4.19) the usual convolution (Cauchy) product of species appearing in [12]. For a general L , the term $L(U \cap V)$ can be considered a measure of the failure of U and V to be disjoint.

5 A combinatorial interpretation

We consider the case where $\mathcal{V} = \text{Set}$ so that $[\mathfrak{S}, \text{Set}]$ is the category of species as studied in [12]. Fix any set Λ . Define the species L by

$$LX = \left\{ S = (S_\lambda)_{\lambda \in \Lambda} \mid S_\lambda \subseteq X, \sum_{\lambda \in \Lambda} S_\lambda = X \right\} \quad (5.21)$$

and $(L\sigma)S = (\sigma S_\lambda)_{\lambda \in \Lambda}$. In other words, a structure of the species L on the set X is a partition of X into a Λ -indexed family of disjoint (possibly empty) subsets.

A structure of the species $F \otimes^L G$ on the set consists of a quintuplet (U, V, S, ϕ, γ) where U, V are subsets of X such that $X = U \cup V$, and S, ϕ, γ are L -, F -, G -structures on $U \cap V$, U , V , respectively.

We write $\#S$ for the cardinality of the set S . We assume Λ is finite and put $\lambda = \#\Lambda$.

The *cardinality sequence* of a species F is the sequence $\#F: \mathbb{N} \rightarrow \mathbb{Z}$ defined by

$$(\#F)(n) = \#F\langle n \rangle.$$

We consider the λ -Hurwitz product (2.2) on $\mathbb{Z}^{\mathbb{N}}$.

Proposition 18. $\#(F \otimes^L G) = \#F \cdot^\lambda \#G$

6 The iterated tensor and coherence

Proposition 19. *An alternative definition of $F \otimes^L G$ is*

$$(F \otimes^L G)X = \sum_{X=A+B+C} L(C) \otimes F(A+C) \otimes G(B+C).$$

Proof. Given $X = A + B + C$, put $U = A + C$ and $V = B + C$. Given $X = U \cup V$, put $A = U \setminus V$, $B = V \setminus U$, and $C = U \cap V$. $\square$

The n -fold version of this tensor product is

$$\begin{aligned} & \otimes_n^L(F_1, \dots, F_n)X \\ &= \sum_{X=\sum_{\emptyset \neq S \subseteq \langle n \rangle} A_S} L\left(\sum_S (\#S - 1) \cdot A_S\right) \otimes F_1\left(\sum_{1 \in S} A_S\right) \otimes \dots \otimes F_n\left(\sum_{n \in S} A_S\right). \end{aligned} \quad (6.22)$$

This yields the formula in Proposition 19 for $n = 2$ by taking $A = A_1, B = A_2, C = A_{\{1,2\}}$. Note that (6.22) is unchanged if we replace $\langle n \rangle$ by any set of cardinality n .

Remark 20. As Joachim Kock reminded me, if we replace $\langle n \rangle$ by the ‘ $(n - 1)$ -simplex’ $[n - 1] = \{0, 1, \dots, n - 1\}$, then the non-empty subsets S correspond to the non-degenerate faces of $[n - 1]$ and $\#S - 1$ is the dimension of the face.

Let us consider the effect of inserting one pair of parentheses in a multiple tensor (6.22). We look at

$$\otimes_{p+1+r}^L (F_1, \dots, F_p, \otimes_q^L (F_{p+1}, \dots, F_{p+q}), F_{p+q+1}, \dots, F_{p+q+r}) X . \quad (6.23)$$

Using (6.22) twice, once with $n = p + 1 + r$ and once with $n = q$, we obtain the expression

$$\begin{aligned} & L \left(\sum_T (\#T - 1) \cdot B_T \right) \otimes F_1 \left(\sum_{1 \in T} B_T \right) \otimes \dots \otimes F_p \left(\sum_{p \in T} B_T \right) \\ & \otimes L \left(\sum_R (\#R - 1) \cdot C_R \right) \otimes F_{p+1} \left(\sum_{p+1 \in R} C_R \right) \otimes \dots \otimes F_{p+q} \left(\sum_{p+q \in R} C_R \right) \\ & \otimes F_{p+q+1} \left(\sum_{p+q+1 \in T} B_T \right) \otimes \dots \otimes F_{p+q+r} \left(\sum_{p+q+r \in T} B_T \right) \end{aligned} \quad (6.24)$$

summed over all families

$$B = (B_T \mid \emptyset \neq T \subseteq \{1, \dots, p, \star, p+q+1, \dots, p+q+r\})$$

providing a partition $X = \sum_T B_T$ of X , together with all families

$$C = (C_R \mid \emptyset \neq R \subseteq \{p+1, \dots, p+q\})$$

providing a partition $\sum_{\star \in T} B_T = \sum_R C_R$ of $\sum_{\star \in T}$. Using the fact that L lands in $\mathcal{ZV}$ and that L is strong monoidal, we obtain the isomorphic expression

$$\begin{aligned} & L \left(\sum_T (\#T - 1) \cdot B_T + \sum_R (\#R - 1) \cdot C_R \right) \\ & \otimes F_1 \left(\sum_{1 \in T} B_T \right) \otimes \dots \otimes F_p \left(\sum_{p \in T} B_T \right) \\ & \otimes F_{p+1} \left(\sum_{p+1 \in R} C_R \right) \otimes \dots \otimes F_{p+q} \left(\sum_{p+q \in R} C_R \right) \\ & \otimes F_{p+q+1} \left(\sum_{p+q+1 \in T} B_T \right) \otimes \dots \otimes F_{p+q+r} \left(\sum_{p+q+r \in T} B_T \right) \end{aligned} \quad (6.25)$$

summed over the same families (B, C) . For $\star \in T$, we have $B_T = \sum_R C_R \cap B_T$. On the other hand, $C_R = \sum_{\star \in T} C_R \cap B_T$. Put

$$Q = \{p+1, \dots, p+q\} \quad \text{and} \quad N = \{1, \dots, p\} \cup \{p+q+1, \dots, p+q+r\} ,$$

and obtain a family

$$A = (A_S \mid \emptyset \neq S \subseteq \langle p + q + r \rangle)$$

partitioning X by defining

$$A_S = \begin{cases} B_S & \text{for } S \cap Q = \emptyset \\ C_{S \cap Q} \cap B_{(S \cap N) \cup \{\star\}} & \text{for } S \cap Q \neq \emptyset . \end{cases}$$

Then we can recover the B and C families via

$$B_T = \begin{cases} A_T & \text{for } \star \notin T \\ \sum_R A_{R \cup (T \setminus \star)} & \text{for } \star \in T \end{cases} \quad \text{and} \quad C_R = \sum_{\star \in T} A_{R \cup (T \setminus \star)} .$$

We have the following equations:

$$(i) \quad \sum_S (\#S - 1) \cdot A_S = \sum_T (\#T - 1) \cdot B_T + \sum_R (\#R - 1) \cdot C_R$$

(ii)

$$\sum_{k \in S} A_S = \begin{cases} \sum_{k \in T} B_T & \text{for } 1 \leq k \leq p \text{ or } p + q + 1 \leq k \leq p + q + r \\ \sum_{k \in R} C_R & \text{for } p + 1 \leq k \leq p + q . \end{cases}$$

This shows that the sum of the expressions (6.25) over the pairs (B, C) is equal to (6.22) with $n = p + q + r$. Remember however that the tensor product $\otimes$ on $\mathfrak{S}$ is not strict symmetric; the symmetry on $\mathfrak{S}$ provides canonical bijections between the left- and right-hand sides of (i) and (ii). Since L is braided, we have constructed a natural isomorphism

$$\begin{aligned} a_{p,q,r}: \otimes_n^L (F_1, \dots, F_{p+q+r}) \\ \cong \otimes_{p+1+r}^L (F_1, \dots, F_p, \otimes_q^L (F_{p+1}, \dots, F_{p+q}), F_{p+q+1}, \dots, F_{p+q+r}) . \end{aligned} \tag{6.26}$$

Now consider the Mac Lane-Stasheff pentagon for 2-fold bracketings of $F_1 \otimes^L F_2 \otimes^L F_3 \otimes^L F_4$ as the vertices. Let $a: H \rightarrow K$ denote one of the edges of the pentagon obtained using the associativity isomorphisms (4.20). There is a composite b of two isomorphisms, each using one instance of an isomorphism (6.26), which goes from $\otimes_4^L (F_1, F_2, F_3, F_4)$ to H , and another one $c: \otimes_4^L (F_1, F_2, F_3, F_4) \rightarrow H$. By coherence of the braided strong monoidal functor L , it follows that $a \circ b = c$. Commutativity of the pentagon is a consequence of commutativity of all these triangular sides of the so-formed pentagonal cone.

7 Promonoidal structures on $\mathfrak{S}$

For finite sets A , B and X , let $\text{Cov}(A, B; X)$ denote the set of jointly surjective pairs (μ, ν) of injective functions

$$A \xrightarrow{\mu} X \xleftarrow{\nu} B .$$

We write $A \times_X B$ for the pullback of μ and ν .

Define a functor

$$P: \mathfrak{S}^{\text{op}} \times \mathfrak{S}^{\text{op}} \times \mathfrak{S} \longrightarrow \mathcal{V}$$

by

$$P(A, B; X) = \sum_{(\mu, \nu) \in \text{Cov}(A, B; X)} L(A \times_X B) . \quad (7.27)$$

Proposition 21. $(F \otimes^L G)X \cong \int^{A, B} P(A, B; X) \otimes FA \otimes GB$

Proof. A universal dinatural transformation

$$\theta_{A, B}: P(A, B; X) \otimes FA \otimes GB \longrightarrow \sum_{X=U \cup V} L(U \cap V) \otimes FU \otimes GV$$

is defined by taking its composite with the injection at $(\mu, \nu) \in \text{Cov}(A, B; X)$ to be obtained from the $(\mu(A), \nu(B))$ injection and the bijections $A \cong \mu(A)$, $B \cong \nu(B)$, $A \times_X B \cong \mu(A) \cap \nu(B)$, noting $X = \mu(A) \cup \nu(B)$. $\square$

By Day's general theory of promonoidal categories [3, 4], we have:

Corollary 22. *If moreover $\mathcal{V}$ is (left and right) closed and sufficiently complete then $\otimes^L$ defines a (left and right) closed monoidal structure on $[\mathfrak{S}, \mathcal{V}]$.*

8 The weighted bimonoidale structure on $\text{fam}\mathfrak{S}$

Consider the 2-category Cat_+ of (small) categories admitting finite coproducts, and finite-coproduct-preserving functors. This becomes a symmetric closed monoidal bicategory (see [5]) with tensor product $\mathcal{A} \boxtimes \mathcal{B}$ representing functors $H: \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{X}$ for which each $H(A, -)$ and each $H(-, B)$ is finite coproduct preserving. Clearly the monoidal category $\mathcal{V}$ of Section 4 is a monoidale (= pseudomonoid) in Cat_+ .

For any category $\mathcal{C}$, we write $\text{fam}\mathcal{C}$ for the free finite coproduct completion of $\mathcal{C}$. That is, fam provides the left biadjoint to the forgetful 2-functor $\text{Cat}_+ \rightarrow \text{Cat}$. Indeed, fam is a strong monoidal pseudofunctor; in particular, there is a canonical equivalence

$$\text{fam}\mathcal{C} \boxtimes \text{fam}\mathcal{D} \simeq \text{fam}(\mathcal{C} \times \mathcal{D}) .$$

Every monoidal category $\mathcal{C}$ determines a monoidale $\text{fam}\mathcal{C}$ in Cat_+ .

Explicitly, the objects of $\text{fam}\mathcal{C}$ can be written formally as $\sum_{s \in S} C_s$ where S is a finite set and $C_s \in \mathcal{C}$. Then, if $\mathcal{C}$ is monoidal, the monoidale structure on $\text{fam}\mathcal{C}$ is defined by

$$\sum_{s \in S} C_s \otimes \sum_{t \in T} D_t = \sum_{(s, t) \in S \times T} C_s \otimes D_t .$$

We are interested in $\text{fam}\mathfrak{S}$. By what we have just said, this is a monoidale in Cat_+ :

$$\sum_{s \in S} U_s \otimes \sum_{t \in T} V_t = \sum_{(s,t) \in S \times T} (U_s + V_t) .$$

Fix a finite set Λ and define $L: \mathfrak{S} \rightarrow \text{Set}$ by $LX = \Lambda^X$ and $L\sigma = \Lambda^{\sigma^{-1}}$. Define a coproduct-preserving functor

$$\Delta: \text{fam}\mathfrak{S} \longrightarrow \text{fam}(\mathfrak{S} \times \mathfrak{S}) \simeq \text{fam}\mathfrak{S} \boxtimes \text{fam}\mathfrak{S} \quad (8.28)$$

by

$$\Delta(X) = \sum_{X=A+B+C} L(C) \cdot (A+C, B+C)$$

for $X \in \mathfrak{S}$.

Proposition 23. *The functor Δ of (8.28) is strong monoidal.*

Proof. In $\Delta(X+Y) = \sum_{X+Y=A+B+C} L(C) \cdot (A+C, B+C)$ we can put

$$P = X \cap A, Q = X \cap B, R = X \cap C, U = Y \cap A, V = Y \cap B, W = Y \cap C$$

to obtain

$$\begin{aligned} \Delta(X+Y) &= \sum_{X=P+Q+R, Y=U+V+W} L(R+W) \cdot (P+U+R+W, Q+V+R+W) \\ &\cong \sum_{X=P+Q+R} L(R) \cdot (P+R, Q+R) \times \sum_{Y=U+V+W} L(W) \cdot (U+W, V+W) \\ &\cong \Delta X \times \Delta Y , \end{aligned}$$

as required. $\square$

9 Weighted categorical derivations

Harking back to Remark 11, we are prompted to consider the 2-category

$$\mathfrak{E} = \text{Hom}(\Sigma\mathfrak{S}, \mathcal{V}\text{-Cat}_{L,+}) . \quad (9.29)$$

Here $\Sigma\mathfrak{S}$ denotes the bicategory with one object (denoted $\star$) whose homcategory is the symmetric groupoid $\mathfrak{S}$; composition is provided by the monoidal structure $+$ on $\mathfrak{S}$. Also $\mathcal{V}\text{-Cat}_{L,+}$ denotes the 2-category of $\mathcal{V}$ -categories admitting finite coproducts and tensoring with the object $L(X)$ of $\mathcal{V}$; the morphisms are $\mathcal{V}$ -functors preserving these colimits; the 2-cells are $\mathcal{V}$ -natural transformations. The objects of (9.29) are pseudofunctors $T: \Sigma\mathfrak{S} \rightarrow \mathcal{V}\text{-Cat}_{L,+}$, the morphisms are pseudonatural transformations, and the 2-cells are modifications (in terminology of [18]). Such

an object T determines a $\mathcal{V}$ -category $T\star = \mathcal{M} \in \mathcal{V}\text{-Cat}_{L,+}$ and a strong monoidal functor $T_{\star\star}: \mathfrak{S} \rightarrow \mathcal{V}\text{-Cat}_{L,+}(\mathcal{M}, \mathcal{M})$. This $T_{\star\star}$ is determined up to equivalence by an endomorphism $D: \mathcal{M} \rightarrow \mathcal{M}$ in $\mathcal{V}\text{-Cat}_{L,+}$ and an involutive Yang-Baxter¹ operator $\rho: D \circ D \Rightarrow D \circ D$ on D (for example, see [14] for terminology). Then $T_{\star\star}\langle n \rangle \cong D^{\circ n}$ and, for the non-identity bijection $\tau: \langle 2 \rangle \rightarrow \langle 2 \rangle$, $T\tau$ transports to ρ . Therefore we shall write the object T of $\mathfrak{E}$ (9.29) as a pair $(\mathcal{M}, D^*)$ where $T\star = \mathcal{M}$ and $T_{\star\star} = D^*$. The morphisms of $\mathfrak{E}$ are then squares

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{K} & \mathcal{N} \\ D^*X \downarrow & \xRightarrow{\kappa_X \cong} & \downarrow E^*X \\ \mathcal{M} & \xrightarrow{K} & \mathcal{N} \end{array}$$

in $\mathcal{V}\text{-Cat}_{L,+}$ which are $\mathcal{V}$ -natural in X and, stacking vertically, respect the tensor in $\mathfrak{S}$. Generalizing the tensor $\boxtimes$ on Cat_+ as in Section 8, we have a tensor, also denoted by $\boxtimes$, on $\mathcal{V}\text{-Cat}_{L,+}$, where the tensor product $\mathcal{A} \boxtimes \mathcal{B}$ represents $\mathcal{V}$ -functors $H: \mathcal{A} \otimes \mathcal{B} \rightarrow \mathcal{X}$ for which each of $H(A, -)$ and $H(-, B)$ preserves finite coproducts and tensoring with each $L(X)$. This makes $\mathcal{V}\text{-Cat}_{L,+}$ into a monoidal bicategory.

This tensor product $\boxtimes$ on $\mathcal{V}\text{-Cat}_{L,+}$ lifts to one, denoted by $\widehat{\boxtimes}$, on $\mathfrak{E}$ (9.29):

$$(\mathcal{M}, D^*) \widehat{\boxtimes} (\mathcal{N}, E^*) = (\mathcal{M} \boxtimes \mathcal{N}, D^* \widehat{\boxtimes} E^*) \quad \text{where} \quad (9.30)$$

$$(D^* \widehat{\boxtimes} E^*)X = \sum_{X=U \cup V} L(U \cap V) \otimes D^*U \boxtimes E^*V. \quad (9.31)$$

To see that $D^* \widehat{\boxtimes} E^*: \mathfrak{S} \rightarrow \mathcal{V}\text{-Cat}_{L,+}(\mathcal{M} \boxtimes \mathcal{N}, \mathcal{M} \boxtimes \mathcal{N})$ is strong monoidal, we calculate:

$$\begin{aligned} & (D^* \widehat{\boxtimes} E^*)(X + Y) \\ & \cong \sum_{X+Y=U \cup V} L(U \cap V) \otimes D^*U \boxtimes E^*V \\ & \cong \sum_{X=U_1 \cup V_1, Y=U_2 \cup V_2} L(U_1 \cap V_1 + U_2 \cap V_2) \otimes (D^*U_1 \circ D^*U_2) \boxtimes (E^*V_1 \circ E^*V_2) \\ & \cong \sum_{X=U_1 \cup V_1, Y=U_2 \cup V_2} L(U_1 \cap V_1) \otimes L(U_2 \cap V_2) \otimes (D^*U_1 \boxtimes E^*V_1) \circ (D^*U_2 \boxtimes E^*V_2) \\ & \cong \sum_{X=U_1 \cup V_1, Y=U_2 \cup V_2} L(U_1 \cap V_1) \otimes (D^*U_1 \boxtimes E^*V_1) \circ L(U_2 \cap V_2) \otimes (D^*U_2 \boxtimes E^*V_2) \\ & \cong (D^* \widehat{\boxtimes} E^*)X \circ (D^* \widehat{\boxtimes} E^*)Y. \end{aligned}$$

¹This is Rodney Baxter <http://en.wikipedia.org/wiki/Rodney_Baxter>, not the author of [1].

In this way, $\mathfrak{E}$ (9.29) becomes a monoidal bicategory.

An L -weighted derivation D^* on a monoidale $\mathcal{M}$ in $\mathcal{V}\text{-Cat}_{L,+}$ is a lifting of the monoidale structure on $\mathcal{M}$ to a monoidale structure on $(\mathcal{M}, D^*)$ in $\mathfrak{E}$ (9.29).

Example 24. An L -weighted derivation $D^*: \mathfrak{S} \rightarrow \mathcal{V}\text{-Cat}_{L,+}([\mathfrak{S}, \mathcal{V}], [\mathfrak{S}, \mathcal{V}])$ on the monoidale $([\mathfrak{S}, \mathcal{V}], \otimes^L)$ is defined by $(D^*X)F = F(X + -)$. The main point is the canonical isomorphism below.

$$\begin{array}{ccc} [\mathfrak{S}, \mathcal{V}] \boxtimes [\mathfrak{S}, \mathcal{V}] & \xrightarrow{\otimes^L} & [\mathfrak{S}, \mathcal{V}] \\ (D^* \boxtimes D^*)X \downarrow & \xRightarrow{\cong} & \downarrow D^*X \\ [\mathfrak{S}, \mathcal{V}] \boxtimes [\mathfrak{S}, \mathcal{V}] & \xrightarrow{\otimes^L} & [\mathfrak{S}, \mathcal{V}] \end{array}$$

Remark 25. The first item of Remark 11 has a categorical version. The forgetful 2-functor $U: \mathfrak{E} \rightarrow \mathcal{V}\text{-Cat}_{L,+}$ has a right biadjoint JS taking the $\mathcal{V}$ -category $\mathcal{A}$ to the object of $\mathfrak{E}$ determined by the $\mathcal{V}$ -category $JS\mathcal{A} = [\mathfrak{S}, \mathcal{A}]$ of species in $\mathcal{A}$, equipped with L -weighted derivation the D^* just as in Example 24 with the codomain $\mathcal{V}$ replaced by $\mathcal{A}$. Since U is strong monoidal, the biadjunction $U \dashv_{\text{bi}} JS$ is monoidal. Consequently the biadjunction lifts to one between the 2-categories of monoidales in $\mathfrak{E}$ and $\mathcal{V}\text{-Cat}_{L,+}$. Indeed U is pseudocomonadic.

10 The iterated tensor product again

Observe the following simple reindexing of (4.19).

Proposition 26. An alternative definition of $F \otimes^L G$ is

$$(F \otimes^L G)X = \sum_{V \subseteq U \subseteq X} L(U \setminus V) \otimes F(U) \otimes G(X \setminus V) .$$

This leads us to another formula for the n -fold L -weighted tensor product. Define the *modified n -filtration set* on any finite set X by:

$$\begin{aligned} \text{mFil}_n X = \{ (U, V) \mid U = (0 = U_0 \subseteq U_1 \subseteq \cdots \subseteq U_{n-1} \subseteq U_n = X), \\ V = (V_0, V_1, \dots, V_{n-1}) \text{ with } V_i \subseteq U_i \text{ for } 0 \leq i < n \} . \end{aligned} \quad (10.32)$$

Proposition 27. An alternative definition of the n -fold tensor product (6.22) is

$$\begin{aligned} \otimes_n^L(F_1, \dots, F_n)X \\ = \sum_{(U, V) \in \text{mFil}_n X} L(U_1 \setminus V_1) \otimes \cdots \otimes L(U_{n-1} \setminus V_{n-1}) \\ \otimes F_1(U_1 \setminus V_0) \otimes \cdots \otimes F_n(U_n \setminus V_{n-1}) . \end{aligned} \quad (10.33)$$

Proof. The formula follows by repeated application of the formula of Proposition 26 in evaluating the left bracketing

$$((\dots(F_1 \otimes^L F_2) \otimes^L \dots) \otimes^L F_n$$

at X . □

Let us relate the formulas (6.22) and (10.33) in the case $n = 3$. A modified 3-filtration $(U, V) \in \text{mFil}_3 X$ of X amounts to subsets $U_1 \subseteq U_2 \subseteq X$ and $V_1 \subseteq U_1, V_2 \subseteq U_2$. With this we can define

$$\begin{aligned} A_1 &= V_1 \cap V_2, \quad A_2 = V_2 \setminus U_1 \cap V_2, \quad A_3 = X \setminus U_2 \\ A_{12} &= U_1 \cap V_2 \setminus A_1, \quad A_{13} = V_1 \setminus A_1, \quad A_{23} = (U_2 \setminus U_1) \setminus A_2, \quad A_{123} = (U_1 \setminus V_1) \setminus A_{12} \end{aligned} \quad (10.34)$$

and verify that $X = A_1 + A_2 + A_3 + A_{12} + A_{13} + A_{23} + A_{123}$. Conversely, given the partition A of X , we can define

$$U_1 = X \setminus (A_2 + A_3 + A_{23}), \quad U_2 = X \setminus A_3, \quad V_1 = A_1 + A_{13}, \quad V_2 = A_1 + A_2 + A_{12}.$$

11 Tensor products for charades

Motivated by Proposition 26, we consider generalizing the tensor product of [16]. Let $\mathfrak{G}_q$ be the groupoid of finite vector spaces over the field $\mathbb{F}_q$ of cardinality q ; the morphisms are linear bijections. We write $V \leq U$ to mean V is an $\mathbb{F}_q$ -linear subspace of U , and we write U/V for the quotient space.

To be specific, take $\mathcal{V} = \text{Vect}_{\mathbb{C}}$ to be the category of complex vector spaces with all linear functions.

Let $L: \mathfrak{G}_q \rightarrow \mathcal{V}$ be a suitable functor: we will consider conditions on it later.

For functors $F, G: \mathfrak{G}_q \rightarrow \mathcal{V}$, define $F \otimes^L G: \mathfrak{G}_q \rightarrow \mathcal{V}$ by

$$(F \otimes^L G)X = \sum_{V \leq U \leq X} L(U/V) \otimes F(U) \otimes G(X/V). \quad (11.35)$$

This leads us to an n -fold tensor product in a manner analogous to (10.33). Define the *modified n -flag set* on any finite $\mathbb{F}_q$ -vector space X by:

$$\begin{aligned} \text{mFlg}_n X &= \{(U, V) \mid U = (0 = U_0 \leq U_1 \leq \dots \leq U_{n-1} \leq U_n = X), \\ &\quad V = (V_0, V_1, \dots, V_{n-1}) \text{ with } V_i \leq U_i \text{ for } 0 \leq i < n\}. \end{aligned} \quad (11.36)$$

Now we put

$$\begin{aligned} &\otimes_n^L (F_1, \dots, F_n)X \\ &= \sum_{(U, V) \in \text{mFlg}_n X} L(U_1 \setminus V_1) \otimes \dots \otimes L(U_{n-1} \setminus V_{n-1}) \\ &\quad \otimes F_1(U_1 \setminus V_0) \otimes \dots \otimes F_n(U_n \setminus V_{n-1}). \end{aligned} \quad (11.37)$$

The formula follows by repeated application of (11.35) in evaluating the left bracketing

$$((\dots(F_1 \otimes^L F_2) \otimes^L \dots) \otimes^L F_n$$

at X .

Let us look at the ternary tensor product

$$\otimes_3^L(F, G, H)X = ((F \otimes^L G) \otimes^L H)X$$

It is a direct sum over modified 3-flags (U, V) on X ; that is, subspaces $U_1 \leq U_2 \leq X$, $V_1 \leq U_1$ and $V_2 \leq U_2$. From these we can uniquely define vector spaces A_S for each $\emptyset \neq S \subseteq \langle 3 \rangle$ via the following diagrams of short exact sequences:

$$\begin{array}{ccccc} A_1 & \twoheadrightarrow & U_1 \cap V_2 & \twoheadrightarrow & A_{12} \\ \downarrow & & \downarrow & & \downarrow \\ V_1 & \twoheadrightarrow & U_1 & \twoheadrightarrow & U_1/V_1 \\ \downarrow & & \downarrow & & \downarrow \\ A_{13} & \twoheadrightarrow & U_1/U_1 \cap V_2 & \twoheadrightarrow & A_{123} \end{array} \quad (11.38)$$

$$\begin{array}{ccccc} U_1 \cap V_2 & \twoheadrightarrow & V_2 & \twoheadrightarrow & A_2 \\ \downarrow & & \downarrow & & \downarrow \\ U_1 & \twoheadrightarrow & U_2 & \twoheadrightarrow & U_2/U_1 \\ \downarrow & & \downarrow & & \downarrow \\ U_1/U_1 \cap U_2 & \twoheadrightarrow & U_2/V_2 & \twoheadrightarrow & A_{23} \end{array} \quad (11.39)$$

$$U_2 \twoheadrightarrow X \twoheadrightarrow A_3, \quad (11.40)$$

from which we see

$$X \cong A_1 \oplus A_2 \oplus A_3 \oplus A_{12} \oplus A_{23} \oplus A_{13} \oplus A_{123}. \quad (11.41)$$

Note also the isomorphisms

$$\begin{aligned} U_1/V_1 &\cong A_{12} \oplus A_{123}, \quad U_2/V_2 \cong A_{13} \oplus A_{23} \oplus A_{123} \\ U_1 &\cong A_1 \oplus A_{12} \oplus A_{13} \oplus A_{123}, \quad U_2/V_1 \cong A_2 \oplus A_{12} \oplus A_{23} \oplus A_{123} \\ X/V_2 &\cong A_3 \oplus A_{13} \oplus A_{23} \oplus A_{123}. \end{aligned} \quad (11.42)$$

On the other hand, we can see that the formula for the right bracketing is

$$\begin{aligned} &(F \otimes^L (G \otimes^L H))X \\ &= \sum_{M \leq N \leq X, M \leq I \leq J \leq X} L(N/M) \otimes L(J/I) \otimes FN \otimes G(J/M) \otimes H(X/I) \end{aligned} \quad (11.43)$$

We can see that this indexing set also leads to a direct sum decomposition (11.41) from the following diagrams of short exact sequences.

$$\begin{array}{ccccc}
A_1 & \twoheadrightarrow & U_1 \cap V_2 & \twoheadrightarrow & A_{12} \\
\downarrow & & \downarrow & & \downarrow \\
V_1 & \twoheadrightarrow & U_1 & \twoheadrightarrow & U_1/V_1 \\
\downarrow & & \downarrow & & \downarrow \\
A_{13} & \twoheadrightarrow & U_1/U_1 \cap V_2 & \twoheadrightarrow & A_{123}
\end{array} \tag{11.44}$$

$$\begin{array}{ccccc}
I \cap N & \twoheadrightarrow & I & \twoheadrightarrow & A_2 \\
\downarrow & & \downarrow & & \downarrow \\
J \cap N & \twoheadrightarrow & J & \twoheadrightarrow & J/J \cap N \\
\downarrow & & \downarrow & & \downarrow \\
A_{123} & \hookrightarrow & J/I & \twoheadrightarrow & A_{23}
\end{array} \tag{11.45}$$

$$J + N \twoheadrightarrow X \twoheadrightarrow A_3, \quad J \cap N \twoheadrightarrow N \twoheadrightarrow A_{13}. \tag{11.46}$$

Note also the isomorphisms

$$\begin{aligned}
N/M &\cong A_{12} \oplus A_{13} \oplus A_{123}, \quad J/I \cong A_{23} \oplus A_{123} \\
N &\cong A_1 \oplus A_{12} \oplus A_{13} \oplus A_{123}, \quad J/M \cong A_2 \oplus A_{12} \oplus A_{23} \oplus A_{123} \\
X/I &\cong A_3 \oplus A_{13} \oplus A_{23} \oplus A_{123}.
\end{aligned} \tag{11.47}$$

In order to have an associativity isomorphism we at least need a canonical isomorphism

$$L(A_{12} \oplus A_{123}) \otimes L(A_{13} \oplus A_{23} \oplus A_{123}) \tag{11.48}$$

$$\cong L(A_{12} \oplus A_{13} \oplus A_{123}) \otimes L(A_{23} \oplus A_{123}). \tag{11.49}$$

We do have such an isomorphism if $L: \mathfrak{G}_q \rightarrow \mathcal{V}$ takes direct sums to tensor products; of course, direct sum of $\mathbb{F}_q$ -vector spaces is neither product nor coproduct in $\mathfrak{G}_q$.

Recall from [16] that $\mathfrak{G}_q$ has a braided promonoidal structure. The convolution structure on $[\mathfrak{G}_q, \mathcal{V}]$ arising from this (as per Day [3, 4]) is precisely the tensor product $F \otimes^J G$ where $JX = I$ for $X = 0$ and $JX = 0$ for $X \neq 0$.

Conjecture. *If L is braided strong promonoidal then (11.35) defines a monoidal structure $\otimes^L$ on $[\mathfrak{G}_q, \mathcal{V}]$.*

Should this be the case, the tensor $\otimes^L$ on $[\mathfrak{G}_q, \mathcal{V}]$ would be obtained from quite an interesting promonoidal structure on $\mathfrak{G}_q$. A short sequence

$$A \xrightarrow{f} X \xrightarrow{g} B \quad (11.50)$$

in $\text{Vect}_{\mathbb{F}_q}$ might be called *short pre-exact* when f is a monomorphism, g is an epimorphism and $\ker f \leq \text{img}$. Write $\text{Spes}(A, B; X)$ for the set of such (f, g) . Put

$$P(A, B; X) = \sum_{(f, g) \in \text{Spes}(A, B; X)} L(\text{im}(g \circ f)) . \quad (11.51)$$

This $P: \mathfrak{G}_q^{\text{op}} \times \mathfrak{G}_q^{\text{op}} \times \mathfrak{G}_q \longrightarrow \mathcal{V}$, defined on morphisms in the obvious way, would give the promonoidal structure in question. The term $L(\text{im}(g \circ f))$ measures the failure of the sequence (11.50) to be exact.

12 The dimension sequence

Following on from Section 11, we take $F \in [\mathfrak{G}_q, \text{Vect}_{\mathbb{C}}]$ and define its *dimension sequence* $\dim F \in \mathbb{Z}^{\mathbb{N}}$ by

$$(\dim F)n = \dim(F(\mathbb{F}_q^n)) . \quad (12.52)$$

This inspires an algebra structure on $A^{\mathbb{N}}$ for any k -algebra A . We assume we have $\lambda \in k$ as before, but also some integer q (not necessarily a prime power). As in [16], we use

$$\phi_n(q) = (q^n - 1)(q^{n-1} - 1) \dots (q - 1) .$$

We define

$$\left[\begin{matrix} n \\ r, s \end{matrix} \right]_q = \frac{\phi_n(q)}{\phi_r(q)\phi_s(q)} , \quad \left[\begin{matrix} n \\ r, s, t \end{matrix} \right]_q = \frac{\phi_n(q)}{\phi_r(q)\phi_s(q)\phi_t(q)} , \dots$$

For $f, g \in A^{\mathbb{N}}$, put

$$f \cdot_q^\lambda g = \sum_{r+s+t=n} \left[\begin{matrix} n \\ r, s, t \end{matrix} \right]_q \lambda^t f(r+t)g(s+t) . \quad (12.53)$$

The calculations of Section 11 show that this is associative at least when $A = \mathbb{Z}$, q is a prime power and $\lambda = \dim L(\mathbb{F})$.

More generally, I claim $A^{\mathbb{N}}$ is an associative k -algebra.

Proposition 28. $\dim(F \otimes^L G) = \dim F \cdot_q^\lambda \dim G$

13 Weighted categorical Rota-Baxter operators

We begin by revisiting the monoidal structure on $\mathbf{Gph}\mathcal{V}$ discussed in Section 3. We now point out that we used $\mathbf{Gph}\mathcal{V}$ there because it is a familiar functor category. However, for the purpose of providing an environment for both weighted derivations and weighted Rota-Baxter operators, there is a full subcategory that will still do the job and is more suggestive of a categorical setting.

Let us write $\mathbf{drb}\mathcal{V}$ for the full subcategory of $\mathbf{Gph}\mathcal{V}$ consisting of the graphs (A, E) which have $E = A$ and have $s, t: A \rightarrow A$ commuting with respect to composition. We emphasise that this is not the functor category $[\Sigma(\mathbb{N} \times \mathbb{N}), \mathcal{V}]$ of objects equipped with a pair of commuting endomorphisms since the morphisms are different. A morphism in $\mathbf{drb}\mathcal{V}$ from (A, s, t) to (B, s, t) consists of a pair (f, ϕ) of morphisms $f, \phi: A \rightarrow B$ such that $s \circ \phi = f \circ s$ and $t \circ \phi = f \circ t$.

The reason $\mathbf{drb}\mathcal{V}$ suffices is that

1. the monoidal structure of Proposition 12 restricts to it,
2. it contains the objects $(A, e, 1_A)$, and
3. it contains the objects $(A, 1_A, e)$.

References

- [1] Glen Earl Baxter, *An analytic problem whose solution follows from a simple algebraic identity*, Pacific Journal Math. **10** (1960) 731–742.
- [2] Pierre Cartier, *On the Structure of Free Baxter Algebras*, Advances in Math. **9** (1972) 253–265.
- [3] Brian J. Day, *Construction of Biclosed Categories* (PhD Thesis, UNSW, 1970) <<http://www.math.mq.edu.au/street/DayPhD.pdf>>.
- [4] Brian J. Day, *On closed categories of functors*, Lecture Notes in Mathematics **137** (Springer-Verlag, 1970) 1–38.
- [5] Brian J. Day and Ross Street, *Monoidal bicategories and Hopf algebroids*, Advances in Math. **129** (1997) 99–157.
- [6] Samuel Eilenberg and G. Max Kelly, *Closed categories*, Proceedings of the Conference on Categorical Algebra (La Jolla, 1965), (Springer-Verlag, 1966) 421–562.
- [7] Peter J. Freyd and G. Max Kelly, *Categories of continuous functors I*, J. Pure and Applied Algebra **2** (1972) 169–191; Erratum Ibid. **4** (1974) 121.

- [8] Mikhail M. Kapranov, *Analogies between the Langlands correspondence and topological quantum field theory*, Progress in Mathematics **131** (Birkhäuser Boston, Massachusetts, 1995) 119–151.
- [9] Li Guo, William Keigher and Shilong Zhang, *Baxter algebras and shuffle products*, Advances in Math. **150** (2000) 117–149.
- [10] Li Guo and William Keigher, *On differential Rota-Baxter algebras*, Journal of Pure and Applied Algebra **212** (2008) 522–540.
- [11] Li Guo, William Keigher and Shilong Zhang, *Monads and distributive laws for Rota-Baxter and differential algebras*, [arXiv:1412.8058v2](https://arxiv.org/abs/1412.8058v2).
- [12] André Joyal, *Une théorie combinatoire des séries formelles*, Advances in Mathematics **42** (1981) 1–82.
- [13] André Joyal, *Foncteurs analytiques et espèces de structures*, Lecture Notes in Mathematics **1234** (Springer 1986) 126–159.
- [14] André Joyal and Ross Street, *Tortile Yang-Baxter operators in tensor categories*, Journal of Pure and Applied Algebra **71** (1991) 43–51.
- [15] André Joyal and Ross Street, *Braided tensor categories*, Advances in Mathematics **102** (1993) 20–78.
- [16] André Joyal and Ross Street, *The category of representations of the general linear groups over a finite field*, Journal of Algebra **176** (1995) 908–946.
- [17] G. Max Kelly, *Basic concepts of enriched category theory*, London Mathematical Society Lecture Note Series **64** (Cambridge University Press, Cambridge, 1982).
- [18] G. Max Kelly and Ross Street, *Review of the elements of 2-categories*, Lecture Notes in Mathematics **420** (Springer-Verlag, 1974) 75–103.
- [19] Saunders Mac Lane, *Categories for the Working Mathematician*, Graduate Texts in Mathematics **5** (Springer-Verlag, 1971).
- [20] Gian-Carlo Rota, *Baxter algebras and combinatorial identities I*, Bulletin Amer. Math. Soc. **5** (1969) 325–329.
- [21] Ross Street, *Monoidal categories in, and linking, geometry and algebra*, Bulletin of the Belgian Mathematical Society – Simon Stevin **19(5)** (2012) 769–821.
- [22] Ross Street, *Skew-closed categories*, Journal of Pure and Applied Algebra **217(6)** (2013) 973–988.